

Nelson Loop

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SKILLS

Languages: JavaScript, TypeScript, C++, C, C#, HTML/CSS, Python, Java, Ruby

Technologies: React, Next.js, Angular, Ruby on Rails, Material-UI, AWS, Docker

EXPERIENCE

Magnet Forensics

Sept 2025 – Dec 2025

Software Developer Co-op | TypeScript, HTML/CSS, Material-UI, C#, Clojure

Waterloo, ON

- Investigated ~ 10 customer-reported production issues during a 2-week support rotation, analyzing logs, debugging services, and coordinating with the support team.
- Led spike planning and managed an epic end-to-end, analyzing requirements, creating actionable tickets, and advising teammates on implementation questions.

Magnet Forensics

Jan 2025 – Apr 2025

Software Developer Co-op | TypeScript, HTML/CSS, Material-UI, C#, Clojure

Waterloo, ON

- Developed and modified RESTful APIs in C# and Clojure, improving backend functionality and reliability.
- Implemented new user-requested features using TypeScript and Material-UI, expanding the product's functionality and enhancing the user experience.

OnOrder

Jan 2024 – Apr 2024

Software Engineering Intern | TypeScript, Next.js, HTML/CSS, Material-UI, Redis, Puppeteer

Toronto, ON

- Integrated OpenAI and HuggingFace APIs to standardize user data, and implemented Redis caching to optimize data retrieval, resulting in reduced API calls by 98% and a faster data normalization process by 85%.
- Employed Puppeteer to systematically extract product details from diverse websites for over 10,000 products, facilitating comprehensive data collection.

Swap Robotics

May 2023 – Aug 2023

Web Developer | JavaScript, HTML/CSS, AWS, Docker, PostgreSQL, Material-UI, Jest

Kitchener, ON

- Shipped 15+ features in a Dockerized React app (JS/HTML/CSS, MUI), including real-time tracking for 20+ robots and fail-safes to halt unsafe autonomous operations.
- Boosted unit test coverage by 70% (Jest) and implemented GitHub Actions on AWS EC2 for automated API test runs on every PR.

Swap Robotics

Sept 2022 – Dec 2022

Autonomous Robotics Software Developer | Python, ROS, Foxglove

Kitchener, ON

- Utilized Python, ROS, and Foxglove to develop code for real-time data tracking, reporting, and visualization of autonomous robots.
- Retrieved and monitored real-time safety diagnostics of a motor MCU from a CAN bus in Python, to enhance safety measures in robot operations.

Tuq Inc.

Jan 2022 – Apr 2022

Software Developer | Ruby, Ruby on Rails, Angular, TypeScript, HTML

Waterloo, ON

- Designed and developed applications based on client specifications, employing Ruby on Rails and Sorbet for backend development, in conjunction with Angular, TypeScript, and HTML for frontend implementation.
- Developed REST APIs that interact with Active Record models, implemented unit tests, and documented with Swagger API for applications, enhancing application reliability and company software quality.

EDUCATION

University of Waterloo

Sept 2021 – Present

BASc. Computer Engineering, Honours, Co-operative Program

Waterloo, ON

- Fourth year undergraduate student.
- Expected Graduation April 2026.